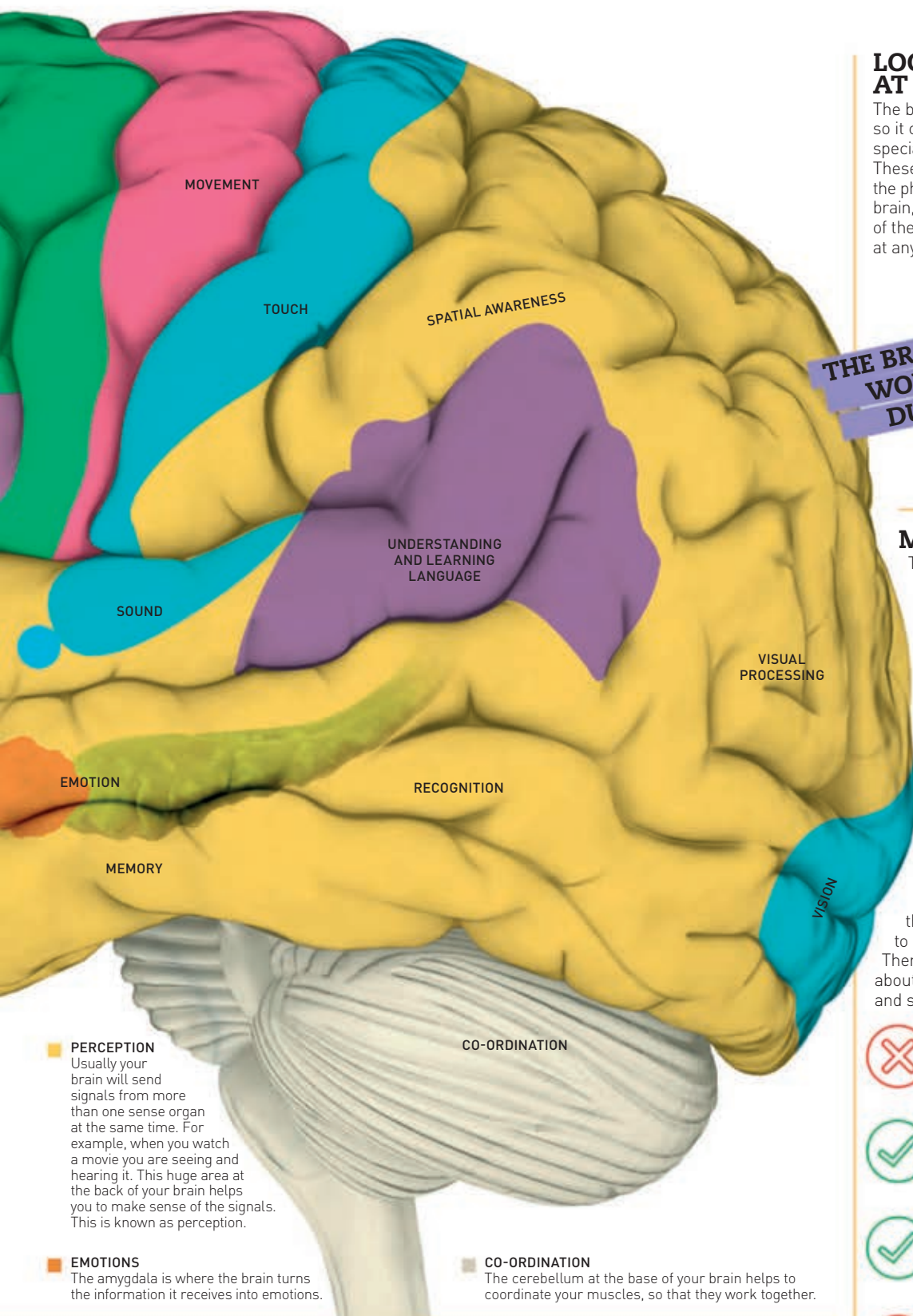




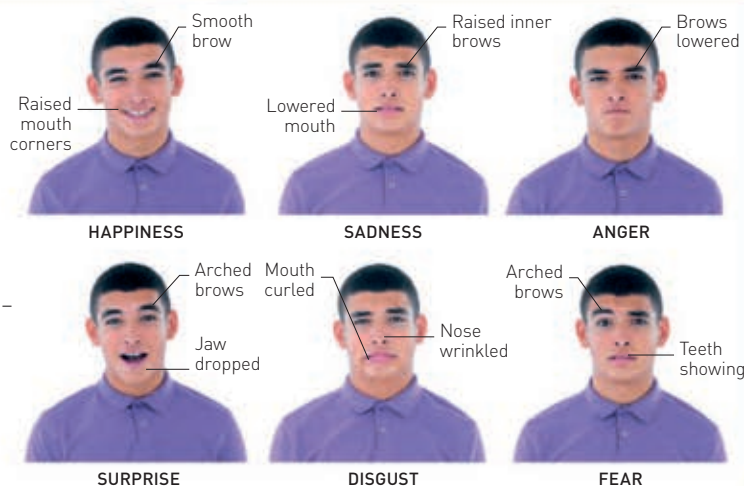
PERCEPTION
Usually your brain will send signals from more than one sense organ at the same time. For example, when you watch a movie you are seeing and hearing it. This huge area at the back of your brain helps you to make sense of the signals. This is known as perception.

EMOTIONS
The amygdala is where the brain turns the information it receives into emotions.

CO-ORDINATION
The cerebellum at the base of your brain helps to coordinate your muscles, so that they work together.

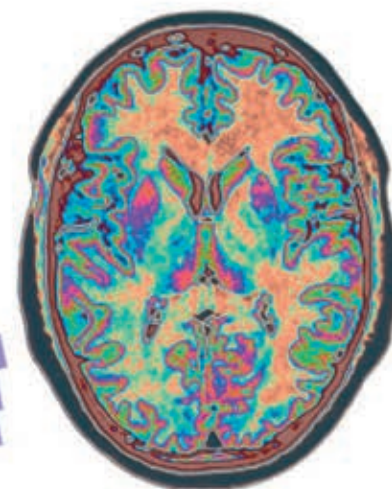
EMOTIONS

The brain processes our feelings. As it does so, signals move through the body so that those feelings become visible to other people. There are six primary emotions, and they all show on the face in a particular way. These facial expressions are the same in everyone – a smile means the same thing whether you live in the Sahara desert or New York City.



LOOKING AT THE BRAIN

The brain is inside the skull, so it can only be seen using special scanning machines. These can be used to show the physical make-up of the brain, or to highlight the parts of the brain that are working at any moment.



MRI SCAN

An MRI scan uses magnetism to produce images of different sections of the brain.

THE BRAIN IS ALWAYS WORKING, EVEN DURING SLEEP

MAKING MEMORIES

The brain absorbs information from the senses, processes all of it into an image or thought, and then stores that image or thought as a memory. Memories can be short-term, such as a phone number you use once, which is held for just as long as you need it, or long-term, such as your first day at school, which you may remember for many years.



TRUE OR FALSE?

The brain is so complex that we are only beginning to understand how it works. There are many popular beliefs about the brain – some are true and some are false.



WE ONLY USE 10 PER CENT OF OUR BRAINS
The truth is that we use all of our brains to complete normal daily tasks.



30,000 NEURONS WOULD FIT ON THE HEAD OF A PIN
This is true, and the brain contains around 100 billion neurons in total.



THE BRAIN DOES NOT FEEL PAIN
The brain does not have pain receptors, so it cannot feel pain.



EINSTEIN'S BRAIN WAS BIGGER THAN AVERAGE
Einstein's brain was a bit smaller than average. Size does not affect intelligence.

SOME UNUSUAL BRAINS

When parts of the brain do not function or function differently, it can affect the way that individuals make sense of the world.

- **AMNESIA**
This is a loss of memory due to a physical or emotional trauma.
- **SYNAESTHESIA**
People with this disorder experience mixed-up senses. For example, some people see colours when they read or hear numbers.
- **DEMENTIA**
This is a set of problems, including difficulties with thinking, memory, problem-solving, and language. It usually affects older adults.
- **OBSESSIVE-COMPULSIVE DISORDER (OCD)**
This is a disorder where people worry about things all the time and repeat actions over and over again.

TRICKS OF THE MIND

Sometimes you cannot believe your eyes – or more accurately, you cannot believe what your brain thinks it is seeing. The brain can be fooled.



WHICH LINE IS LONGER?

Do you see one of these lines as longer? This is a visual illusion – the lines are both the same length.



WHICH IS BIGGER?

The red dot on the right looks bigger, but it is not. Your brain judges it in relation to the blue dots around it.